Internet & Web Technology – HTML Tags and Details

1. HTML: Basics of HTML

What is HTML?

HTML (HyperText Markup Language) is the standard markup language used to create the structure of web pages.

HTML uses tags to tell the browser how content should be displayed.

Basic HTML Structure

```
<!DOCTYPE html>
<html>
<head>
  <title>My First Web Page</title>
</head>

<body>
  <h1>Welcome to HTML</h1>
  <p>This is my first web page.</p>
</body>
</html>
```

Important Basic Tags

Tag	Purpose
`<!DOCTYPE html>`	Defines HTML5 document
`<html>`	Root element
`<head>`	Contains page information
`<title>`	Sets browser-tab title
`<body>`	Contains visible content
`<h1>` to `<h6>`	Headings
`<p>`	Paragraph
` `	Line break
`<hr>`	Horizontal line

Heading Example

<h1>Main Heading</h1>
<h2>Sub Heading</h2>
<h3>Small Heading</h3>
<h1> is the largest/most important heading, while <h6> is the smallest.

Paragraph

```
<p>This is a paragraph.</p>
```

Line Break

```
Hello<br>

World
```

Output:

```
Hello

World
```

Horizontal Line

```
<hr>
```

---

2. HTML: Basics of HTML (2)

HTML Elements

An HTML element normally consists of:

```
<tag>Content</tag>
```

Example:

```
<p>Hello Students</p>
```

Here:

 → Opening tag

Hello Students → Content

 → Closing tag

HTML Attributes

Attributes provide additional information about an element.

```
<p id="para1">Hello Students</p>
```

Here `id="para1"` is an attribute.

Another example:

```
<img src="photo.jpg" alt="Student Photo">
```

Attributes:

`src` → Image location

`alt` → Alternative text

Common Attributes

Attribute	Use
`id`	**Unique identification**
`class`	**Groups elements**
`style`	**Inline CSS**
`title`	**Tooltip information**
`src`	**Source of image/media**
`href`	**Link destination**
`alt`	**Alternative text**
`width`	**Width**
`height`	**Height**

---

3. Formatting and Fonts

HTML provides several tags for formatting text.

Bold

```
<b>This is bold</b>
```

Strong

```
<strong>Important Text</strong>
```

`<strong>` gives semantic importance to the text.

Italic

```
<i>This is italic</i>
```

Emphasis

```
<em>This is emphasized</em>
```

Underline

```
<u>This is underlined</u>
```

Small Text

```
<small>This is small text</small>
```

Highlight Text

```
<mark>Important Text</mark>
```

Deleted Text

```
<del>Old Price</del>
```

Inserted Text

```
<ins>New Text</ins>
```

Subscript

```
H<sub>2</sub>O
```

Output:

H_2O

Superscript

```
X<sup>2</sup>
```

Output:

X²

Font using CSS

The old `<font>` tag is obsolete in HTML5. Use CSS instead:

```
<p style="font-family: Arial; font-size: 20px;">
    Hello Students
</p>
```

---

4. Commenting Code, Color and Hyperlink

HTML Comments

Comments are not displayed in the browser.

```
<!-- This is an HTML comment -->
```

Example:

```
<!-- Student Information -->
<p>Salil</p>
```

Comments are useful for explaining code.

Colors

Colors can be applied using CSS.

Color Name

```
<p style="color: red;">Hello</p>
```

Hexadecimal Color

```
<p style="color: #ff0000;">Hello</p>
```

Background Color

```
<body style="background-color: lightblue;">
```

Text and Background

```
<p style="color: white; background-color: blue;">
    Welcome to HTML
</p>
```

---

5. Hyperlink

The <a> tag is used to create hyperlinks.

Basic Syntax

```
<a href="https://www.google.com">Google</a>
```

Open Link in New Tab

```
<a href="https://www.google.com" target="_blank">
    Open Google
</a>
```

Important attributes:

Attribute	Purpose
Href	**Specifies URL**
target	**Specifies where to open the link**
title	**Provides additional information**

Email Link

```
<a href="mailto:student@example.com">
    Send Email
</a>
```

---

6. Lists

HTML provides three major types of lists.

Ordered List

Uses numbers.

```
<ol>
    <li>HTML</li>
    <li>CSS</li>
    <li>JavaScript</li>
</ol>
```

Output:

HTML

CSS

JavaScript

Unordered List

Uses bullets.

```
<ul>
    <li>HTML</li>
    <li>CSS</li>
    <li>JavaScript</li>
</ul>
```

Description List

```
<dl>

    <dt>HTML</dt>

    <dd>HyperText Markup Language</dd>


    <dt>CSS</dt>

    <dd>Cascading Style Sheets</dd>

</dl>
```

Tags:

`<dl>` → Description list

`<dt>` → Description term

`<dd>` → Description details

---

7. Tables

The `<table>` tag is used to create tables.

Basic Table

```
<table border="1">
  <tr>
    <th>Name</th>
    <th>Course</th>
  </tr>

  <tr>
    <td>Rahul</td>
    <td>CSE</td>
  </tr>

  <tr>
    <td>Amit</td>
    <td>IT</td>
  </tr>
</table>
```

Important Table Tags

Tag	Purpose
`<table>`	Creates table
`<tr>`	Table row
`<th>`	Table heading
`<td>`	Table data
`<caption>`	Table title
`<thead>`	Header section
`<tbody>`	Body section
`<tfoot>`	Footer section

Colspan

Combines columns.

```
<td colspan="2">Total</td>
```

Rowspan

Combines rows.

```
<td rowspan="2">CSE</td>
```

---

8. Images

The `<img>` tag is used to display an image.

```
<img src="student.jpg" alt="Student Image">
```

Image with Width and Height

```
<img src="student.jpg"
     alt="Student Image"
```

```
     width="300"

     height="200">
```

Important attributes:

Attribute	Purpose
`src`	**Image path**
`alt`	**Alternative text**
`width`	**Image width**
`height`	**Image height**

`<img>` is a void element, so it does not have a closing `</img>` tag.

---

9. Forms

Forms are used to collect information from users.

Common applications:

Registration

Login

Feedback

Contact forms

Student information

Basic Form

```
<form>
  <label>Name:</label>
  <input type="text">

  <br><br>

  <label>Email:</label>
  <input type="email">
```

```
  <br><br>

  <input type="submit" value="Submit">
</form>
```

Important Form Tags

Tag	Purpose
`<form>`	Creates form
`<label>`	Label for input
`<input>`	Input field
`<textarea>`	Multi-line input
`<select>`	Dropdown
`<option>`	Dropdown option
`<button>`	Button
`<fieldset>`	Groups form controls
`<legend>`	Caption for fieldset

Input Types

```
<input type="text">
<input type="password">
<input type="email">
<input type="number">
<input type="date">
<input type="radio">
<input type="checkbox">
<input type="file">
<input type="submit">
<input type="reset">
```

Radio Button

```
<input type="radio" name="gender"> Male

<input type="radio" name="gender"> Female
```

Checkbox

```
<input type="checkbox"> C

<input type="checkbox"> C++

<input type="checkbox"> Java
```

Dropdown

```
<select>

    <option>CSE</option>

    <option>IT</option>

    <option>ECE</option>

</select>
```

Textarea

```
<textarea rows="5" cols="30"></textarea>
```

---

10. XHTML

XHTML = Extensible HyperText Markup Language

XHTML is HTML written according to stricter XML rules.

HTML

```
<P>Hello</P>
```

XHTML

```
<p>Hello</p>
```

Important XHTML rules:

Tags should be properly nested.

Tags should be closed.

Elements should be written consistently.

Attribute values should be quoted.

Elements should follow XML-style syntax.

Example:

```
<img src="photo.jpg" alt="Photo" />
```

---

11. Meta Tags

The `<meta>` tag provides information about an HTML document.

Meta tags are placed inside `<head>`.

Character Encoding

```
<meta charset="UTF-8">
```

Description

```
<meta name="description"
      content="Internet and Web Technology">
```

Keywords

```
<meta name="keywords"
      content="HTML, CSS, JavaScript">
```

Author

```
<meta name="author"
      content="Salil Bhatnagar">
```

Responsive Web Page

```
<meta name="viewport"
      content="width=device-width, initial-scale=1.0">
```

The viewport meta tag is especially important for responsive websites.

---

12. Character Entities

HTML entities are used to display reserved or special characters.

Entity	Display
<	<
>	>
&	&
"	"
'	'
	Non-breaking space
©	©
®	®
™	™

Example:

```
<p>5 &lt; 10</p>

<p>Tom &amp; Jerry</p>

<p>&copy; 2026</p>
```

---

13. Frames and Framesets

Older HTML used `<frameset>` and `<frame>` to divide the browser window into multiple sections.

Example:

```
<frameset cols="30%,70%">
  <frame src="menu.html">
  <frame src="home.html">
</frameset>
```

Important Tags

Tag	Purpose
`<frameset>`	Defines frame layout
`<frame>`	Defines individual frame

Important Note

`<frameset>` and `<frame>` are obsolete and not supported in HTML5.

For modern web pages, use:

```
<iframe src="page.html"></iframe>
```

The `<iframe>` element embeds another page/document inside the current page.

Example:

```
<iframe
    src="https://example.com"
    width="600"
    height="400">
</iframe>
```

---

14. Browser Architecture

A browser is software used to access and display web pages.

Examples include:

Chrome
Firefox
Edge
Safari

Basic Browser Processing

User
↓
Browser
↓

Enter URL
↓
DNS
↓
Web Server
↓
HTTP/HTTPS Request
↓
HTML/CSS/JavaScript Response
↓
Browser Rendering
↓
Web Page

Main Browser Components

Component	Function
User Interface	**Browser buttons, address bar, tabs**
Browser Engine	**Coordinates browser operations**
Rendering Engine	**Displays HTML/CSS**
JavaScript Engine	**Executes JavaScript**
Networking	**Handles HTTP/HTTPS**
UI Backend	**Draws basic UI components**
Data Storage	**Stores cookies, cache, local storage**

---

15. Website Structure

A website normally contains different types of files.

Example:

```
mywebsite/
│
├── index.html
├── about.html
├── contact.html
│
├── css/
│   └── style.css
│
├── js/
│   └── script.js
│
└── images/
    └── logo.png
```

index.html

Usually the main/home page.

CSS

Used for design and presentation.

```
css/style.css
```

JavaScript

Used for functionality and interaction.

```
js/script.js
```

Images

Stored separately:

```
images/logo.png
```

---

16. Overview and Features of HTML5

HTML5 is the modern version of HTML.

It introduced semantic elements, multimedia support, improved forms, graphics, and APIs.

Important HTML5 Semantic Tags

<header>
<nav>
<main>
<section>
<article>
<aside>
<footer>

Example

```
<!DOCTYPE html>
<html>
<head>
  <title>HTML5 Website</title>
</head>

<body>

<header>
  <h1>My Website</h1>
</header>

<nav>
  <a href="#">Home</a>
  <a href="#">About</a>
</nav>

<main>
  <section>
    <h2>Welcome</h2>
    <p>This is my website.</p>
  </section>

  <article>
    <h2>HTML5</h2>
    <p>HTML5 provides modern web features.</p>
  </article>
</main>

<footer>
  <p>&copy; 2026 My Website</p>
</footer>

</body>
</html>
```

---

17. HTML5 Multimedia

Audio

```
<audio controls>
    <source src="song.mp3" type="audio/mpeg">
</audio>
```

Video

```
<video width="500" controls>
    <source src="video.mp4" type="video/mp4">
</video>
```

YouTube/External Content

Usually embedded using `<iframe>`:

```
<iframe
    width="560"
    height="315"
    src="https://www.youtube.com/embed/VIDEO_ID">
</iframe>
```

---

18. HTML5 Form Features

HTML5 introduced new input types.

```
<input type="email">
<input type="url">
<input type="number">
<input type="date">
<input type="time">
<input type="month">
<input type="week">
<input type="color">
<input type="range">
<input type="search">
```

Required Field

```
<input type="text" required>
```

Placeholder

```
<input type="text" placeholder="Enter your name">
```

Minimum and Maximum

```
<input type="number" min="1" max="100">
```

---

19. HTML5 Graphics

Canvas

The `<canvas>` element provides a drawing area.

```
<canvas id="myCanvas"
        width="300"
        height="150">
</canvas>
```

JavaScript is normally used to draw on the canvas.

SVG

SVG is used for scalable vector graphics.

---

20. Important HTML Tags – Quick Revision

Category	Important Tags
Basic	**<html>, <head>, <title>, <body>**
Headings	**<h1> – <h6>**
Text	**<p>, , <hr>**
Formatting	**<b>, , <i>, , <u>, <mark>, <small>, <del>, <ins>**
Links	**<a>**

Category	Important Tags
Images	`<img>`
Lists	`<ol>`, `<ul>`, `<li>`, `<dl>`, `<dt>`, `<dd>`
Tables	`<table>`, `<tr>`, `<th>`, `<td>`
Forms	`<form>`, `<input>`, `<label>`, `<select>`, `<option>`, `<textarea>`, `<button>`
Frames	`<iframe>`
HTML5 Semantic	`<header>`, `<nav>`, `<main>`, `<section>`, `<article>`, `<aside>`, `<footer>`
Multimedia	`<audio>`, `<video>`, `<source>`
Graphics	`<canvas>`, `<svg>`
Metadata	`<meta>`, `<link>`
Structure	`<div>`, `<span>`

Complete Learning Sequence

For students, I recommend teaching this unit in this order:

HTML Introduction
↓
Basic Structure & Elements
↓
Headings & Paragraphs
↓
Formatting & Fonts
↓
Comments & Colors
↓
Hyperlinks
↓
Lists
↓
Tables
↓
Images
↓
Forms
↓
XHTML
↓
Meta Tags & Character Entities
↓
Frames / Framesets & iframe
↓
Browser Architecture
↓

Website Structure
↓
HTML5 Semantic Tags
↓
Audio & Video
↓
HTML5 Forms
↓
Canvas & SVG